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$$\int \frac{x^5}{x^3 - 1} dx$$

$$\text{Stel } t = x^3 - 1 \Rightarrow dt = 3x^2 dx$$

$$= \int \frac{t+1}{3t} dt$$

$$= \frac{1}{3} \int \frac{t+1}{t} dt$$

$$= \frac{1}{3} \int dt + \frac{1}{3} \int \frac{dt}{t}$$

$$= \frac{1}{3} t + \frac{1}{3} \ln |t| + C$$

$$= \frac{x^3 - 1 + \ln |x^3 - 1|}{3} + C$$

References